

Preparatory Course for MSc Students, WiSe 2026/27

Exercise Sheets for Section 5 (Analysis and Topology) of the lecture notes

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How to use these sheets. Each sheet corresponds to one section of the lecture notes. Exercises marked **(W)** are warm-ups meant to be relatively simpler; unmarked exercises form the core of the session; exercises marked **(*)** are more challenging. Notation follows the lecture notes; $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$. The numbering continues the sheets for Sections 1–4, so cross-references such as “Exercise 4.8” refer to those sheets. As on Sheet 3, Hermitian inner products are antilinear in the first argument; on ℓ^2 this means $\langle a, b \rangle = \sum_n \overline{a_n} b_n$.

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Sheet 6: Topological and Metric Spaces

Exercise 6.1 (W) *Topologies on finite sets.*

- (a) Let $X = \{a, b, c\}$. Which of the following are topologies on X ?

$$\begin{aligned}\mathcal{T}_1 &= \{\emptyset, X, \{a\}\}, & \mathcal{T}_2 &= \{\emptyset, X, \{a\}, \{b\}\}, \\ \mathcal{T}_3 &= \{\emptyset, X, \{a\}, \{a, b\}\}, & \mathcal{T}_4 &= \{\emptyset, X, \{a, b\}, \{b, c\}\}.\end{aligned}$$

- (b) List all topologies on $X = \{0, 1\}$. Which of them are Hausdorff?
(c) Show that every Hausdorff topology on a finite set is discrete.
(d) What is the topology on $\{a, b, c\}$ generated by $\mathcal{S} = \{\{a, b\}, \{b, c\}\}$?

Exercise 6.2 (W) *Open and closed sets.*

- (a) Show that arbitrary intersections and finite unions of closed sets are closed. Give an example showing that infinite unions of closed sets need not be closed.
(b) Decide whether the following subsets of $\mathbb{R}$ (with the Euclidean topology) are open, closed, both or neither: $[0, 1)$, $\mathbb{Z}$, $\mathbb{Q}$, $\{\frac{1}{n} \mid n \in \mathbb{N}\}$, $\{\frac{1}{n} \mid n \in \mathbb{N}\} \cup \{0\}$, $\mathbb{R}$.
(c) In which topology on a set X is every subset both open and closed? Which subsets are closed in the trivial topology?

Exercise 6.3 *Hausdorff spaces and limits.*

- (a) Prove that limits of sequences in a Hausdorff space are unique.
(b) Let X carry the trivial topology and have at least two points. Show that every sequence converges to every point.
(c) Let $\mathbb{N}$ carry the cofinite topology. Show that it is not Hausdorff and that the sequence $x_n = n$ converges to *every* point of $\mathbb{N}$.
(d) Show that every metric space is Hausdorff.

Exercise 6.4 *Continuous maps.*

- (a) Show that compositions of continuous maps are continuous, and that f is continuous iff preimages of closed sets are closed.
(b) Show that every map out of a discrete space is continuous, and that every map into a space with the trivial topology is continuous.
(c) Show that $\text{id} : (\mathbb{R}, \mathcal{T}_{\text{discrete}}) \rightarrow (\mathbb{R}, \mathcal{T}_{\text{eucl}})$ is continuous but its inverse is not.
(d) Show that the Sorgenfrey line $\mathbb{R}_l$ is Hausdorff, and that $\text{id} : \mathbb{R}_{\text{eucl}} \rightarrow \mathbb{R}_l$ is not continuous.
(e) Show that $f : \mathbb{R} \rightarrow \mathbb{R}$, $f = \mathbf{1}_{\mathbb{Q}}$, is discontinuous at every point (use sequences).
(f) Let $f(x, y) = \frac{xy}{x^2 + y^2}$ for $(x, y) \neq 0$ and $f(0, 0) = 0$. Show that $t \mapsto f(t, y_0)$ and $t \mapsto f(x_0, t)$ are continuous for all x_0, y_0 , but that f is not continuous at 0.

Exercise 6.5 *Metrics and their topologies.*

- (a) The *discrete metric* on a set X is $d(x, y) = 1$ for $x \neq y$ and $d(x, x) = 0$. Show that it induces the discrete topology. Compute $B_1(x)$ and the closed ball $\{y \mid d(x, y) \leq 1\}$. Is the closed ball the closure of the open ball?
- (b) Let d be a metric. Show that $d' = \frac{d}{1+d}$ is a metric inducing the same topology as d , and that d' is bounded. *Hint:* $\varphi(t) = \frac{t}{1+t}$ is increasing and satisfies $\varphi(a+b) \leq \varphi(a) + \varphi(b)$.
- (c) Show that in a metric space a set A is closed iff it contains the limits of all convergent sequences in A .

Exercise 6.6 *Norms on $\mathbb{R}^n$.*

- (a) Show that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms on $\mathbb{R}^n$. Sketch the unit balls of $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty$ in $\mathbb{R}^2$.
- (b) Show that $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2 \leq n \|x\|_\infty$. Conclude directly that these three norms are equivalent.
- (c) Show that for $0 < p < 1$ the formula for $\|x\|_p$ does *not* define a norm on $\mathbb{R}^2$.
- (d) Show that a norm coming from an inner product satisfies the parallelogram law $\|v+w\|^2 + \|v-w\|^2 = 2\|v\|^2 + 2\|w\|^2$. Deduce that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $\mathbb{R}^2$ do not come from inner products.
- (e) Show that $p(f) = |f(0)|$ is a seminorm, but not a norm, on $C^0([0, 1])$.

Exercise 6.7 *Compactness.*

- (a) Show (in any topological space) that a finite union of compact sets is compact and that a closed subset of a compact space is compact.
- (b) Prove Theorem 5.11 directly from the definition of compactness. Deduce the extreme value theorem: a continuous $f : K \rightarrow \mathbb{R}$ on a non-empty compact space attains its maximum and minimum.
- (c) Show that $(0, 1]$ is not compact using the sequence criterion, and that $\{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$ is compact using the open cover definition.
- (d) Show that a space with the discrete topology is compact iff it is finite.
- (e) Using Heine–Borel on $\text{Mat}(n \times n, \mathbb{R}) \cong \mathbb{R}^{n^2}$, decide which of $O(n)$, $SO(n)$, $SL(n, \mathbb{R})$ ($n \geq 2$) and $GL(n, \mathbb{R})$ are compact.
- (f) (*) Show that a compact subset of a Hausdorff space is closed.

Exercise 6.8 (*) *All norms on $\mathbb{R}^n$ are equivalent.* Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$.

- (a) Show that $\|x\| \leq C\|x\|_2$ with $C = (\sum_i \|e_i\|^2)^{1/2}$. Deduce that $x \mapsto \|x\|$ is continuous with respect to $\|\cdot\|_2$.
- (b) Using compactness of the Euclidean unit sphere, show that there is $m > 0$ with $m\|x\|_2 \leq \|x\|$.
- (c) Conclude that all norms on a finite-dimensional vector space are equivalent, and that every finite-dimensional normed space is complete.

Sheet 7: Normed, Banach and Hilbert Spaces; Bounded Operators

Exercise 7.1 (W) *Operator norms of matrices.* Let $A = (a_{ij}) \in \text{Mat}(n \times n, \mathbb{R})$ act on $\mathbb{R}^n$.

- (a) Show that with respect to $\|\cdot\|_\infty$, $\|A\|_{\text{op}} = \max_i \sum_j |a_{ij}|$ (maximal row sum).
- (b) Show that with respect to $\|\cdot\|_1$, $\|A\|_{\text{op}} = \max_j \sum_i |a_{ij}|$ (maximal column sum).
- (c) Show that with respect to $\|\cdot\|_2$, $\|A\|_{\text{op}} = \sqrt{\lambda_{\max}(A^t A)}$. *Hint:* Exercise 4.8(b).
- (d) Compute all three operator norms for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 7.2 *Bounded and unbounded operators.*

- (a) Show that $D = \frac{d}{dx} : (\mathbb{R}[x], \|\cdot\|_\infty) \rightarrow (\mathbb{R}[x], \|\cdot\|_\infty)$, with $\|p\|_\infty = \sup_{[0,1]} |p|$, is not bounded.
- (b) Show that $I : (C^0([0,1]), \|\cdot\|_\infty) \rightarrow (C^0([0,1]), \|\cdot\|_\infty)$, $(If)(x) = \int_0^x f(t) dt$, is bounded, and compute $\|I\|_{\text{op}}$.
- (c) Show that $\text{ev}_0(f) = f(0)$ is a bounded linear functional on $(C^0([0,1]), \|\cdot\|_\infty)$ but not on $(C^0([0,1]), \|\cdot\|_1)$. What does this say about the dependence of the continuous dual space on the norm?
- (d) For $g \in C^0([0,1])$ show that $M_g : f \mapsto gf$ satisfies $\|M_g\|_{\text{op}} = \|g\|_\infty$ on $(C^0([0,1]), \|\cdot\|_\infty)$.
- (e) On ℓ^2 let $S(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$ and $L(a_1, a_2, \dots) = (a_2, a_3, \dots)$. Show that $\|Sa\|_2 = \|a\|_2$, that $\|L\|_{\text{op}} = 1$, and that $LS = \text{id} \neq SL$. Compare with Exercise 1.2(e) and Exercise 2.3(c).

Exercise 7.3 *The space of bounded operators.* Let V, W be normed spaces.

- (a) Show that $\|A\|_{\text{op}} = \sup_{\|v\|=1} \|Av\| = \sup_{\|v\| \leq 1} \|Av\|$ and $\|Av\| \leq \|A\|_{\text{op}} \|v\|$.
- (b) Show that $\|\cdot\|_{\text{op}}$ is a norm on $L(V, W)$ and that $\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}$.
- (c) (*) Show that $L(V, W)$ is a Banach space whenever W is. In particular, the continuous dual $V^* = L(V, \mathbb{K})$ is always complete.

Exercise 7.4 *Completeness.*

- (a) Show that $\mathbb{Q}$ with $d(x, y) = |x - y|$ is not complete.
- (b) Show that a closed subset of a complete metric space is complete, and that a complete subset of a metric space is closed.
- (c) Let $c_{00} \subset \ell^2$ be the subspace of sequences with only finitely many non-zero terms. Show that c_{00} is not complete, but that it is dense in ℓ^2 .
- (d) Let $f_n(x) = \sqrt{(x - \frac{1}{2})^2 + \frac{1}{n}}$ on $[0, 1]$. Show that $f_n \rightarrow |x - \frac{1}{2}|$ uniformly. Deduce that $(C^1([0, 1]), \|\cdot\|_\infty)$ is not complete.
- (e) (*) Show that $C^1([0, 1])$ is complete with respect to $\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty$.

Exercise 7.5 *The sequence spaces ℓ^p and Hilbert spaces.*

- (a) Show that $\ell^1 \subsetneq \ell^2 \subsetneq \ell^\infty$ and $\|a\|_\infty \leq \|a\|_2 \leq \|a\|_1$.
- (b) Show that the norm $\|\cdot\|_\infty$ on ℓ^∞ does not come from an inner product.

- (c) Let (e_k) be the standard unit vectors in ℓ^2 . Show that (e_k) is a bounded sequence without a convergent subsequence. Why does this not contradict Heine–Borel?
- (d) (Bessel's inequality.) Let $(u_k)_{k \in \mathbb{N}}$ be an orthonormal sequence in a Hilbert space H . Show that $\sum_k |\langle u_k, v \rangle|^2 \leq \|v\|^2$ for all $v \in H$. *Hint:* expand $\|v - \sum_{k \leq N} \langle u_k, v \rangle u_k\|^2 \geq 0$.
- (e) For $b \in \ell^2$ show that $\alpha_b(a) = \langle b, a \rangle$ defines an element of $(\ell^2)^*$ with $\|\alpha_b\|_{\text{op}} = \|b\|_2$.

Exercise 7.6 *The Banach fixed point theorem.* Let (X, d) be a non-empty complete metric space and $f : X \rightarrow X$ with $d(f(x), f(y)) \leq q d(x, y)$, $q \in [0, 1)$.

- (a) Complete the proof sketched in the lecture. Show that $d(x_{n+1}, x_n) \leq q^n d(x_1, x_0)$ and $d(x_m, x_n) \leq \frac{q^n}{1-q} d(x_1, x_0)$ for $m > n$. Deduce that (x_n) converges, that its limit is a fixed point, that the fixed point is unique, and the error estimate $d(x_n, x^*) \leq \frac{q^n}{1-q} d(x_1, x_0)$.
- (b) Show that $\cos$ has a unique fixed point in $[0, 1]$.
- (c) Show that $f(x) = \frac{x}{2} + \frac{1}{x}$ is a contraction of $[1, 2]$ into itself. What is its fixed point? How many iterations from $x_0 = 1$ guarantee an error below 10^{-6} according to (a)?
- (d) Show that the hypotheses cannot be dropped: (i) $f(x) = \frac{x}{2}$ on $(0, 1]$; (ii) $f(x) = x + \frac{1}{x}$ on $[1, \infty)$, which satisfies $|f(x) - f(y)| < |x - y|$ for $x \neq y$.
- (e) (*) Let $|\lambda| < 1$. Show that $Tf(x) = 1 + \lambda \int_0^x f(t) dt$ is a contraction of $(C^0([0, 1]), \|\cdot\|_\infty)$. Identify its fixed point and the iterates starting from $f_0 = 0$.

Exercise 7.7 *Neumann series.* Let V be a Banach space and $A \in L(V)$ with $\|A\|_{\text{op}} < 1$.

- (a) Show that $\sum_k A^k$ converges in $L(V)$, that its limit is $(\text{id} - A)^{-1}$, and that $\|(\text{id} - A)^{-1}\|_{\text{op}} \leq \frac{1}{1 - \|A\|_{\text{op}}}$.
- (b) Deduce that $\text{GL}(n, \mathbb{R})$ is open in $\text{Mat}(n \times n, \mathbb{R})$. More precisely, if A is invertible and $\|B - A\|_{\text{op}} < \|A^{-1}\|_{\text{op}}^{-1}$, then B is invertible.
- (c) Solve $f(x) - \frac{1}{2} \int_0^1 xt f(t) dt = x$ on $C^0([0, 1])$ by summing the Neumann series. Check your answer.

Exercise 7.8 *The matrix exponential and linear ODEs.*

- (a) Compute $\exp(tA)$ for $A = \text{diag}(\lambda_1, \dots, \lambda_n)$, for $A = J_{\lambda, 2}$, and for $A = J_{0, 3}$.
- (b) Compute $\exp(tJ)$ for $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- (c) Solve $\dot{x} = Ax$, $x(0) = (1, 0)^t$ for $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. *Hint:* Exercise 4.3(a).
- (d) Show that $\exp(A + B) \neq \exp(A)\exp(B)$ for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.
- (e) Show that $\frac{\exp(hA) - I}{h} \rightarrow A$ as $h \rightarrow 0$ in operator norm, and deduce that $\frac{d}{dt} \exp(tA) = A \exp(tA)$. Conclude that $x(t) = \exp(tA)x_0$ solves $\dot{x} = Ax$, $x(0) = x_0$.
- (f) (*) Show that $\det \exp(A) = e^{\text{tr} A}$ for $A \in \text{Mat}(n \times n, \mathbb{C})$.
- (g) Show that $\exp(A) \in SO(n)$ if A is skew-symmetric. *Hint:* for the determinant, use (f).

Exercise 7.9 *Topological groups and topological vector spaces.*

- (a) Show that every normed space is a topological vector space (Example 5.15).

- (b) Show that $\text{GL}(n, \mathbb{R}) \subset \mathbb{R}^{n^2}$, with the subspace topology, is a topological group. *Hint:* Cramer's rule.
- (c) Let G be a topological group and $g \in G$. Show that left translation $L_g(h) = gh$ is a homeomorphism. Deduce that the topology is determined by the neighbourhoods of the identity.
- (d) On $C^0(\mathbb{R})$ consider the seminorms $p_k(f) = \sup_{[-k, k]} |f|$, $k \in \mathbb{N}$. Show that $f_m \rightarrow f$ in the resulting locally convex topology iff $f_m \rightarrow f$ uniformly on every compact subset. Show that this topology is Hausdorff.

Sheet 8: Test Functions and Distributions

Exercise 8.1 (W) *Test functions.* Let $\psi(x) = e^{-1/x}$ for $x > 0$ and $\psi(x) = 0$ for $x \leq 0$.

- (a) Show by induction that for $x > 0$, $\psi^{(k)}(x) = p_k(1/x) e^{-1/x}$ for a polynomial p_k . Deduce that $\psi \in C^\infty(\mathbb{R})$. *Hint:* $\lim_{x \searrow 0} x^{-m} e^{-1/x} = 0$ for every m .
- (b) Show that $\varphi(x) = \psi(1-x^2)$ lies in $C_c^\infty(\mathbb{R})$ with $\text{supp } \varphi = [-1, 1]$, and that $\varphi > 0$ on $(-1, 1)$.
- (c) Show that $C_c^\infty(\mathbb{R}^n)$ is a vector space that is closed under ∂^α and under multiplication by functions in $C^\infty(\mathbb{R}^n)$.
- (d) Show that a polynomial with compact support is zero. (So test functions are never given by power series on all of $\mathbb{R}$.)

Exercise 8.2 *Convergence in $C_c^\infty(\mathbb{R})$.* Let $\varphi \in C_c^\infty(\mathbb{R})$ be as in Exercise 8.1(b). Decide which sequences converge to 0 in $C_c^\infty(\mathbb{R})$:

$$(a) \frac{1}{m} \varphi(x), \quad (b) \frac{1}{m} \varphi(mx), \quad (c) \frac{1}{m} \varphi(x-m), \quad (d) e^{-m} \varphi(mx).$$

For (c), find a linear functional T on $C_c^\infty(\mathbb{R})$ with $T(\varphi_m) \not\rightarrow 0$ that is nevertheless a distribution.

Exercise 8.3 *Examples of distributions.* Show that the following are distributions on $\mathbb{R}$:

- (a) $T(\varphi) = \varphi'(0)$;
- (b) the *Dirac comb* $C(\varphi) = \sum_{k \in \mathbb{Z}} \varphi(k)$;
- (c) T_H for the Heaviside function $H = \mathbf{1}_{[0, \infty)}$;
- (d) (*) the principal value $\text{p.v.} \frac{1}{x}(\varphi) = \lim_{\epsilon \searrow 0} \int_{|x| > \epsilon} \frac{\varphi(x)}{x} dx$. *Hint:* if $\text{supp } \varphi \subseteq [-R, R]$, write the integral as $\int_{\epsilon < |x| < R} \frac{\varphi(x) - \varphi(0)}{x} dx$.
- (e) (*) Show that δ_0 is not regular, i.e. not of the form T_f with $f \in L^1_{\text{loc}}$.

Exercise 8.4 *Derivatives of distributions.*

- (a) Show that $\dot{T}$ is again a distribution, and that $\dot{T}_f = T_{f'}$ for $f \in C^1(\mathbb{R})$. (This justifies the sign in the definition.)
- (b) Compute $\dot{T}_H$ for the Heaviside function.
- (c) Compute the first and second distributional derivatives of $T_{|x|}$.
- (d) Compute $\dot{\delta}_0$, and show that $x \delta_0 = 0$ and $x \dot{\delta}_0 = -\delta_0$.
- (e) Show the Leibniz rule $(\psi T)^\cdot = \dot{\psi} T + \psi \dot{T}$ for $\psi \in C^\infty(\mathbb{R})$.
- (f) (*) Show that $\frac{d}{dx} T_{\log|x|} = \text{p.v.} \frac{1}{x}$.

Exercise 8.5 *Approximating the delta distribution.* Say that $T_m \rightarrow T$ in $\mathcal{D}'(\mathbb{R})$ if $T_m(\varphi) \rightarrow T(\varphi)$ for every $\varphi \in C_c^\infty(\mathbb{R})$.

- (a) Let $f_m = \frac{m}{2} \mathbf{1}_{[-1/m, 1/m]}$. Show that $T_{f_m} \rightarrow \delta_0$.
- (b) More generally, let $\rho \in L^1(\mathbb{R})$ with $\int \rho = 1$, $\rho \geq 0$, $\rho = 0$ outside $[-1, 1]$, and $\rho_m(x) = m\rho(mx)$. Show that $T_{\rho_m} \rightarrow \delta_0$.
- (c) Show that $T_{\sin(mx)} \rightarrow 0$, although $\sin(mx)$ does not converge pointwise.

- (d) Show that $T_m \rightarrow T$ implies $\dot{T}_m \rightarrow \dot{T}$. Deduce that $T_{\rho'_m} \rightarrow \dot{\delta}_0$ for ρ as in (b) with $\rho \in C^1$.

Exercise 8.6 *Convolution and fundamental solutions in one dimension.*

- (a) For continuous f , show that $(T_f * \varphi)(x) = \int f(y)\varphi(x-y) dy$, the usual convolution.
- (b) Compute $\delta_a * \varphi$ and $\dot{\delta}_0 * \varphi$.
- (c) Show that the Heaviside function is a fundamental solution of $\frac{d}{dx}$, and that $E(x) = \max(x, 0)$ and $\frac{1}{2}|x|$ are fundamental solutions of $\frac{d^2}{dx^2}$. Why are fundamental solutions not unique?
- (d) For $f \in C_c^\infty(\mathbb{R})$, compute $u = E * f$ with $E(x) = \max(x, 0)$ and verify directly that $u'' = f$.
- (e) Find a fundamental solution of $L = \frac{d}{dx} - \lambda$ (for $\lambda \in \mathbb{R}$) of the form $E = H(x)g(x)$. Use it to solve $u' - \lambda u = f$ for $f \in C_c^\infty(\mathbb{R})$.

Exercise 8.7 (*) *The Newton potential.* Let $n \geq 2$, $r = \|x\|$ and N as in Example 5.22.

- (a) Show that $N \in L_{\text{loc}}^1(\mathbb{R}^n)$, so that T_N is a distribution.
- (b) Show that for a radial function $u(x) = g(r)$ on $\mathbb{R}^n \setminus \{0\}$, $\Delta u = g''(r) + \frac{n-1}{r}g'(r)$. Deduce that $\Delta N = 0$ on $\mathbb{R}^n \setminus \{0\}$.
- (c) Check that formally setting $n = 1$, $\omega_1 = 2$ in the formula for N gives the fundamental solution $\frac{1}{2}|x|$ from Exercise 8.6(c).
- (d) Show that $\Delta T_N = \delta_0$. *Hint:* write $T_N(\Delta\varphi) = \lim_{\epsilon \rightarrow 0} \int_{\|x\| > \epsilon} N \Delta\varphi$ and use Green's identity $\int_\Omega (N \Delta\varphi - \varphi \Delta N) = \int_{\partial\Omega} (N \partial_\nu \varphi - \varphi \partial_\nu N) dS$, together with $\nabla N(x) = \frac{x}{\omega_n \|x\|^n}$.

Sheet 9: Differential Calculus in Normed Spaces

Exercise 9.1 (W) *Partial derivatives, gradients and Jacobi matrices.*

- (a) Let $f(x, y) = x^2y + \sin(xy)$. Compute ∇f , the directional derivative of f at $(1, 0)$ in the direction $v = (\frac{3}{5}, \frac{4}{5})$, and the unit direction of steepest ascent at $(1, 0)$.
- (b) Compute the Jacobi matrix and its determinant for polar coordinates $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$.
- (c) Compute the Jacobi matrix of $f(x, y, z) = (xyz, x^2 + z, e^y)$ at $(1, 0, 2)$.

Exercise 9.2 *Partial, directional and total differentiability.*

- (a) Let $f(x, y) = \frac{xy}{x^2+y^2}$, $f(0, 0) = 0$. Show that both partial derivatives exist at 0, but f is not totally differentiable there.
- (b) Let $g(x, y) = \frac{x^2y}{x^2+y^2}$, $g(0, 0) = 0$. Show that g is continuous and that *all* directional derivatives $\partial_v g(0)$ exist, but g is not totally differentiable at 0. *Hint:* is $v \mapsto \partial_v g(0)$ linear?
- (c) (*) Let $h(x) = \|x\|^2 \sin(1/\|x\|)$ for $x \neq 0$ and $h(0) = 0$ on $\mathbb{R}^2$. Show that h is totally differentiable at 0, but its partial derivatives are not continuous at 0. So the converse of Theorem 5.26 fails.
- (c) Show that total differentiability at x implies continuity at x , and that $\partial_v f(x) = df_x(v)$.

Exercise 9.3 *Derivatives of maps between normed spaces.* Compute the total derivative in each case, directly from Definition 5.25.

- (a) A bounded linear map $A \in L(V, W)$.
- (b) A bounded bilinear map $\beta : V \times W \rightarrow Z$ (with $\|\beta(v, w)\| \leq C\|v\|\|w\|$), at (v, w) .
- (c) $f : \text{Mat}(n \times n, \mathbb{R}) \rightarrow \text{Mat}(n \times n, \mathbb{R})$, $f(X) = X^2$.
- (d) The quadratic form $q(x) = x^t A x$ on $\mathbb{R}^n$. Show that $\nabla q(x) = (A + A^t)x$.
- (e) $\text{inv} : \text{GL}(n, \mathbb{R}) \rightarrow \text{GL}(n, \mathbb{R})$, $X \mapsto X^{-1}$. *Hint:* Neumann series.
- (f) $\det : \text{Mat}(n \times n, \mathbb{R}) \rightarrow \mathbb{R}$ at I and at an invertible X .
- (g) $E : (C^0([0, 1]), \|\cdot\|_\infty) \rightarrow \mathbb{R}$, $E(f) = \int_0^1 f(t)^2 dt$.

Exercise 9.4 *The chain rule.*

- (a) For differentiable $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ let $\tilde{u} = u \circ \Phi$ with Φ as in Exercise 9.1(b). Express $\partial_r \tilde{u}$ and $\partial_\theta \tilde{u}$ in terms of $\partial_x u$, $\partial_y u$.
- (b) Let $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{R}^n$ be a differentiable curve and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable. Show $\frac{d}{dt} f(\gamma(t)) = \langle \nabla f(\gamma(t)), \gamma'(t) \rangle$. Deduce that $\nabla f(p)$ is orthogonal to $\gamma'(0)$ for every curve in the level set $\{f = f(p)\}$ through p .
- (c) (Euler's identity.) Let $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ be differentiable and homogeneous of degree k , i.e. $f(\lambda x) = \lambda^k f(x)$ for $\lambda > 0$. Show that $\sum_i x_i \partial_i f(x) = k f(x)$.
- (d) Use the chain rule and Exercise 9.3(b) to derive the product rule $\frac{d}{dt} \langle x(t), y(t) \rangle = \langle \dot{x}, y \rangle + \langle x, \dot{y} \rangle$ for curves in $\mathbb{R}^n$.
- (e) Use the chain rule and Exercise 9.3(f) to show $\frac{d}{dt} \det \exp(tA) = \text{tr}(A) \det \exp(tA)$. Deduce $\det \exp(A) = e^{\text{tr} A}$ once more.

(f) (*) Show that in polar coordinates $\Delta u = \partial_r^2 \tilde{u} + \frac{1}{r} \partial_r \tilde{u} + \frac{1}{r^2} \partial_\theta^2 \tilde{u}$.

Exercise 9.5 *Gradients with respect to bilinear forms; Hamiltonian mechanics.* Let B be a non-degenerate bilinear form on $\mathbb{R}^n$ and define $\nabla^B f$ by $df_x(v) = B(\nabla^B f(x), v)$ for all v , as in Remark 5.34.

- (a) Show that $\nabla^B f(x)$ exists and is unique. If G is the Gram matrix of B , show that $\nabla^B f = (G^t)^{-1} \nabla f$.
- (b) Compute $\nabla^\eta f$ for the Lorentz inner product η on $\mathbb{R}^4$.
- (c) Let $n = 2m$ and Ω the symplectic form with Gram matrix $J = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$ in coordinates (q, p) . Compute $\nabla^\Omega f$ carefully, and check whether Hamilton's equations $\dot{q} = \partial_p H$, $\dot{p} = -\partial_q H$ can be written as $(\dot{q}, \dot{p}) = \nabla^\Omega H$. Compare your sign with the formula in Remark 5.34.
- (d) Show that H is constant along every solution of Hamilton's equations. *Hint:* skew-symmetry of Ω .
- (e) For the harmonic oscillator $H(q, p) = \frac{1}{2}(p^2 + q^2)$ on $\mathbb{R}^2$, write Hamilton's equations as $\dot{x} = Ax$ and solve them using Exercise 7.8(b).

Exercise 9.6 *The inverse function theorem.*

- (a) Show that polar coordinates $\Phi : (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}^2 \setminus \{0\}$ are a local diffeomorphism near every point, but not injective. What does this say about the inverse function theorem?
- (b) Let $f(x, y) = (x^2 - y^2, 2xy)$ (that is, $z \mapsto z^2$ on $\mathbb{C}$). At which points does the inverse function theorem apply? Compute the derivative of the local inverse g near $f(1, 1) = (0, 2)$ at the point $(0, 2)$.
- (c) Show that $f(x) = x^3$ is a smooth bijection $\mathbb{R} \rightarrow \mathbb{R}$ but not a diffeomorphism. Which hypothesis of Theorem 5.35 fails?
- (d) Show that if f is a diffeomorphism near x_0 with inverse g , then $dg_{f(x_0)} = (df_{x_0})^{-1}$. In particular, invertibility of df_{x_0} is necessary for a differentiable local inverse.

Exercise 9.7 *The implicit function theorem.*

- (a) Let $f(x, y) = x^2 + y^2 - 1$. Near which points of the circle $\{f = 0\}$ can it be written as a graph $y = h(x)$? Compute $h'(x)$ by implicit differentiation.
- (b) Differentiate $f(x, h(x)) = 0$ to show $dh_x = -(\partial_y f)^{-1} \partial_x f$ (evaluated at $(x, h(x))$).
- (c) (Folium of Descartes.) Show that near $(\frac{3}{2}, \frac{3}{2})$ the curve $x^3 + y^3 - 3xy = 0$ is the graph of a smooth function $y = h(x)$, and compute $h'(\frac{3}{2})$. Why does the theorem not apply at $(0, 0)$?
- (d) Show that near $(1, 1, 1)$ the solution set of $x^2 + y^2 + z^2 = 3$, $x - y = 0$ can be written as $(y, z) = h(x)$. Compute $h'(1)$ using (b), and check your answer by solving explicitly.
- (e) Show that near $(1, 1, 1)$ the system $x^2 + y^2 + z^2 = 3$, $x + y + z = 3$ cannot be solved in this way. Interpret geometrically.
- (f) (*) (Simple roots depend smoothly on coefficients.) Let p_0 be a real polynomial of degree n with a simple real root y_0 . Show that every polynomial p of degree $\leq n$ whose coefficients are close to those of p_0 has a unique root near y_0 , depending smoothly on the coefficients. Deduce that a simple real eigenvalue of a real matrix depends smoothly on the matrix entries.